

Compute average baud rate and minimum bandwidth requirement for 2B1Q, if data Rate is 44 Kbps and case factor, $C = 0.5$

Ans.

For 2B1Q

$$r = \frac{\text{Data Element}}{\text{Signal Element}}$$
$$= \frac{2}{1} = 2$$

$$\text{Baud rate, } S = \frac{C \times N}{r}$$
$$= \frac{0.5 \times 44 \times 10^3}{2}$$
$$= 11,000 \text{ baud}$$
$$(11 \text{ kbaud})$$

Minimum Bandwidth,

$$B_{\min} = \frac{N}{4}$$
$$= 11 \text{ KHz}$$